

Start off Reviewing the Roadmap

Last time: Can Alice authenticate to the domain?

Today's question: After Alice authenticates, what is she actually authorized to access?

VMware tools have been verified.

If we want to go onto the local account on the client, we type in CLIENT01/adminpc

To verify the domain secure channel, use the command **Test-ComputerSecureChannel -Verbose**. It should output True.

This command checks whether your Windows computer still has a valid trust relationship with the AD domain. When a computer joins AD, a computer account is created. The trust relationship is maintained using that user account.

Another command: **nltest /sc_verify:kevinlab.local**

Client01-DomainJoined SnapShot taken.

The term: "Auditing your existing permission structure," we are referring to reviewing who currently has access to what, and deciding whether that access is actually appropriate.

Run a command: **Get-ChildItem "C:\Shares\Departments"**

To verify Domain Local Groups: **Get-ADGroupMember DL_IT_Share_Modify**

GG_IT_Admins -> DL_IT_Share_Modify (Within All: IT Users)

To verify user membership: **Get-ADGroupMember GG_HR_Users**

Alice HR -> ahr -> GG_HR_Users -> DL_HR_Share_Modify -> HR Folder

AGDLP: stands for Accounts, Global Groups, Domain Local Groups, Permissions

Finish NTFS permissions. (C:\Shares\Departments\)

Let's give each DL group the ability to modify, read and execute, list folder contents, read, and write. The whole idea is a security concept. A bad design would be to give Alice permissions directly within the HR folder.

Next, we will create the SMB shares. So far, I have only completed the NTFS permissions. But, we need to make the resources accessible across the network.

In order to see where everything is shared currently, run: **Get-SmbShare**. SMB shares are folders, printers, or other resources that a Windows computer makes available over the network using the SMB protocol. To create new SMB Shares:

New-SmbShare ``-Name "NAME" ` -Path "PATH" ` -Change Access "..."` ``-FullAccess "..."`

ChangeAccess specifies users or groups that receive ChangeSMB share permissions.

FullAccess gives users/groups Full Control over the SMB share.

Permission layers: NTFS and Share

- Share permission: controls access when somebody access the resource over the network (example \\SRV-DC01\HR)
- NTFS: controls access to Windows file/folder itself

When accessing the network, both matter. User -> SMB Share Permission -> NTFS Permission -> Resource. The user's effective network access is constrained by both layers.

I want to prove this authorization works. Use CLIENT01 as ahr. Open File Explorer and Enter \\SRV-DC01\HR. Access works! Put inside: alice-test.txt, inside write: Alice can modify the HR share. Open it again, and delete.

Now, trying something she doesn't have access to. (\\SRV-DC01\Finance). Denied! Popup saying ask network administrator for access.

User	Resource	Expected	Actual
Alice HR	HR	Allowed	Allowed
Alice HR	Finance	Denied	Denied
Frank Finance	Finance	Allowed	Allowed
Frank Finance	HR	Denied	Denied

Testing a second user, Frank Finance! Test \\SRV-DC01\Finance...Allowed! Test \\SRV-DC01\HR...Denied!

Now, moving onto our First Security-Focused GPO.

Server Manager -> Tools -> Group Policy Management

Domains, kevinlab.local...Right-click: KEVINLAB-Computers or Workstations OU

Choose: Create a GPO in this domain, and link it here...Name:

GPO-Workstation-Security-Auditing

Edit the Policy directly: Policies -> Windows Settings -> Security Settings -> Advanced Audit

Policy Configuration -> Audit Policies -> Detailed Tracking -> Audit Process Creation.

Enable audit process creation and select success.

Windows can log process creation as: Security Event ID 4688. So when processes start on CLIENT01, we'll have security telemetry showing that activity.

To apply and verify: go on CLIENT01 with adminpc and run **gpupdate /force**

This command tells Windows to immediately re-process Group Policy settings instead of waiting for the normal automatic refresh. Run **gpresult /r**

I can see GPO-Workstation-Security-Auditing under Applied Group Policy Objects.

Lets run a few processes (Notepad, calc, whoami). If we open Event Viewer, go to Windows Logs -> Security...we can filter for Event ID 4688 and we can see the logs for starting these processes.